OLYMPIC SPORTS PERFORMANCE ANALYSIS

DOCUMENT 03

Visualization and Dashboard

Dashboard purpose, visual components, website integration, and testing

Academic Project Documentation

Purpose and Scope

This document records the project's approach based on the project context available. Compare all descriptions and counts with the final Tableau workbook and dataset before academic submission.

Data note: Important: project context describes the workbook as using demonstration/synthetic data. Treat findings as findings from this dataset, not verified historical Olympic facts, unless independently verified.

1. Purpose of the Dashboard

The Tableau dashboard is designed to make multi-dimensional sports-performance information easier to explore. Users can compare countries, years, regions, and sport categories through visual summaries rather than separate tables.

2. Dashboard Views

- Global medal distribution map — geographic overview of medal outcomes.
- Annual medal trends — shows selected medal measures over time.
- Country performance metrics — supports country comparisons.
- Average medals per athlete — relates medal outcomes to a participation measure, depending on the workbook formula.
- Regional efficiency pie chart — summarizes the workbook's regional-efficiency metric.
- Investment-versus-medals bubble chart — compares investment values and medal outcomes in the demonstration dataset.
- Host-country comparison — compares host results with a reference measure.
- Regional sport-category donut chart — shows sport-category composition for a selected region or metric.

3. Dashboard Organization

Project context identifies two main themes: “Global Sports Performance Analytics” and “Olympic Performance and Regional Efficiency.” Use the exact titles shown in the published workbook when finalizing this report. The Flask site has separate home, dashboard, and story routes.

4. Interactivity

Where configured in Tableau, filters and selections can focus the view on a country, year, region, or sport category. Only claim an interaction exists if it is present and working in the published workbook. Embedded dashboards also depend on sharing and privacy settings.

5. Design Principles

- Use clear titles and meaningful axis labels.
- Keep number formats and category labels consistent.
- Use color consistently and provide legends.
- Show units, time period, and metric definitions.
- Avoid crowded labels and unnecessary decoration.
- Check readability at common laptop and mobile widths.

6. Website Integration

The Flask app defines `/` to render `index.html`, `/dashboard` to render `dashboard.html`, and `/story` to render `story.html`. Static assets are stored under `static/assets/`. The deployed app uses Gunicorn on Render.

7. Testing Checklist

- Open the home page and test navigation links.
- Open the dashboard page and confirm the Tableau view loads.
- Open the story page and confirm its content loads.
- Test filters and tooltips if included.
- Check desktop and mobile layouts.
- Confirm the Tableau workbook is viewable by the intended audience.

8. Accessibility

Use readable text, descriptive links, and sufficient contrast. Add a short explanation of each chart and do not rely on color as the only way to distinguish categories.